Adaptive Check-In Workflows for Conversational Mental Health Companions: Scoping Review and Research Agenda

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Abstract

Background. Conversational artificial intelligence (AI) for mental health has matured rapidly. Randomized controlled trials of cognitive behavioral chatbots have produced clinically measurable symptom reductions; generative-AI chatbots have produced large-effect improvements on clinical outcomes (Cohen $d = 0.79$–0.90); and meta-analyses of app-supported smartphone interventions for mental health show small-to-moderate effects (Hedges $g \approx 0.28$–0.58). Despite this progress, no peer-reviewed deployment combines a *learned, longitudinal model of the client*, a *theory-grounded scheduling policy* that selects when to reach out and what to offer, a *multi-framework intervention library* authored by the subject matter expert (SME) and selected from by the policy, and a *crisis-safe conversational substrate*—all evaluated on listener-centered outcomes.

Objective. This scoping review synthesizes the literatures of just-in-time adaptive interventions (JITAIs), micro-randomized trials (MRTs), conversational mental health agents, and the digital therapeutic alliance (DTA) to assess whether an integrated, adaptive check-in workflow can be assembled today using existing methods, and to specify a research agenda for evaluating one.

Methods. We followed the Arksey–O'Malley five-stage framework with the methodological refinements of Levac et al., and report findings according to the PRISMA Extension for Scoping Reviews (PRISMA-ScR). Twelve Boolean queries spanning three thematic pillars (*Sense*, *Decide*, *Act*) and six cross-cutting concerns (therapeutic frameworks, wellbeing techniques, crisis safety, engagement, ethics, evaluation methodology) were executed in April 2026 against PubMed/MEDLINE, PsycINFO (via Ovid), and Scopus, with citation-context verification through the Scite verified-DOI service for every source retained for synthesis. We applied prespecified inclusion criteria (verified DOI; peer-reviewed venue or recent foundational preprint with ≥ 5 citing publications; topical relevance to one of eight research questions; sufficient methodological detail) and synthesized findings narratively across the three pillars and six cross-cutting concerns.

Results. We synthesized 62 verified-DOI sources. The *Sense* pillar offers a mature methodological chain from ecological momentary assessment through passive sensing and digital phenotyping to large-language-model (LLM) state inference. The *Decide* pillar provides the most developed methodological stack: a JITAI conceptual framework, MRT evaluation design with statistical machinery, contextual-bandit and reinforcement-learning policy classes with both online and offline precedent, and one mental-health-relevant adaptive-messaging RCT (DIAMANTE). The *Act* pillar contributes RCT-validated chatbot precedents (Woebot, Wysa, Therabot), recent meta-analytic effects on the order of $g \approx 0.28$–0.64 across depressive and anxious symptoms, an

attrition baseline of $\sim$22% for chatbot RCTs, and an evolving DTA literature. Cross-cutting findings cover therapeutic-framework evidence bases, wellbeing-technique mechanisms, crisis-safety architectures, engagement and attrition baselines, ethical constraints, and evaluation instruments. We define five operational integration criteria (C1–C5) and find no peer-reviewed system that demonstrably satisfies all five jointly. A subsystem-by-subsystem feasibility analysis indicates that every component required to fill the gap appears available from existing methods, instruments, and currently available models.

Conclusions. The integration of a learned, theory-grounded, scheduled check-in workflow into a multi-framework conversational mental health companion appears methodologically tractable today but has not been attempted in peer-reviewed deployment. We propose a *Decide-first* prototype evaluated by an MRT against a fixed-cadence baseline, with proximal-effect, alliance, and retention reporting. The literature supports the design; the prototype, the trial, and the clinical or behavioral outcomes are explicitly future work.

Keywords. mHealth; mental health; conversational agent; chatbot; large language model; just-in-time adaptive intervention; micro-randomized trial; scoping review; digital therapeutic alliance; behavior change.

1 Introduction

1.1 Background

Conversational artificial intelligence (AI) for mental health has made rapid empirical progress. The Woebot RCT of Fitzpatrick et al. [1], the Wysa real-world evaluation of Inkster et al. [2], and the *NEJM AI* Therabot RCT of Heinz et al. [3] collectively demonstrate that chat-substrate interventions can produce clinically meaningful symptom reductions and therapist-comparable alliance ratings. The meta-analytic baseline across 66 smartphone-intervention RCTs in the Linardon et al. synthesis [4] is $g \approx 0.28$–0.58 across depression, anxiety, stress, quality-of-life, and social-anxiety outcomes, with cognitive behavioral therapy (CBT)-based apps with engagement support producing the largest effects. This empirical base is sufficient to justify continued deployment of supportive conversational companions; it is *not* sufficient to claim the field has found the right shape for them.

1.2 Prior Work and Persistent Gaps

Three deeper gaps persist across the published corpus. (a) *No longitudinal learned model of the client.* Most chatbot deployments rely on static intake forms or implicit context windows; they do not learn what this user is like over weeks or months in ways that can transfer across sessions. (b) *No theory-grounded scheduling policy.* Delivery schedules are typically fixed (daily check-in, weekly session) or entirely user-initiated; the extensive literature on just-in-time adaptive interventions (JITAIs), micro-randomized trials (MRTs), and contextual-bandit / reinforcement-learning (RL) policies developed in mobile-health biostatistics [5–7] has been applied to physical-activity outcomes and, in one recent study [8], to depression-relevant populations via text messaging, but has not been integrated into a multi-framework conversational mental health companion. (c) *No multi-framework intervention library bound to a scheduling policy.* Most chatbots commit to a single therapeutic framework at design time; the subject matter expert (SME) who knows the audience has no seam through which to swap in acceptance and commitment therapy (ACT), motivational interviewing (MI), dialectical behavior therapy (DBT) skills, or compassion-focused therapy (CFT), and no scheduling intelligence that learns which move to select for which state.

1.3 Objective and Research Questions

The objective of this scoping review is twofold: (1) to synthesize the literatures that bear on those three gaps and (2) to ask whether they can be filled with currently available methods. We organize the field around three pillars and six cross-cutting concerns:

- *Sense* (Section 3.2): how can a system build a longitudinal, psychologically meaningful model of a client from the conversation itself (and, optionally, light-touch ecological momentary assessment, EMA)?
- *Decide* (Section 3.3): when and what should the system deliver—how does the field connect JITAI/MRT/contextual-bandit policy mathematics to conversational mental health content?
- *Act* (Section 3.4): how are chat-delivered interventions performing empirically, and what does the digital therapeutic alliance (DTA) literature say about the mediator between delivery and outcomes?
- Cross-cutting concerns: therapeutic frameworks (Section 3.5); wellbeing techniques (Section 3.6); crisis detection and safety (Section 3.7); engagement and attrition (Section 3.8); ethics, regulation, equity, and privacy (Section 3.9); and evaluation methodology (Section 3.10).

A future implementation need not be limited to language alone: opt-in wearable signals (sleep, activity, heart-rate variability, temperature, recovery proxies) [9, 10] can provide a lightweight sensing branch into the *Sense* layer that helps the *Decide* policy time low-risk interventions (grounding, paced breathing, reflective check-ins) at moments of heightened stress, fatigue, or dysregulation, without establishing a diagnosis on their own.

The review answers eight research questions, summarized in Table 1. The synthesis is illustrated throughout against one candidate architecture (named in Section 4.4) that exemplifies each design commitment in concrete form; that architecture is presented as an exemplar, not as the unit of evaluation.

Scope of the present paper. This is a scoping literature review and research agenda, not a system or evaluation paper. The candidate architecture referenced in the Discussion currently realizes only the conversational delivery layer with a baseline keyword crisis stack; the *Sense* layer, the *Decide* scheduling policy, the micro-randomized trial, and any clinical or behavioral outcomes that this review motivates are explicitly positioned as future work (Section 4.4). No human subjects have been recruited and no proximal or distal outcomes are reported in the present paper.

2 Methods

2.1 Study Design and Reporting Standards

This is a scoping review that adopts the Arksey–O'Malley five-stage framework [11] with the methodological refinements of Levac et al. [12], and follows the PRISMA Extension for Scoping Reviews (PRISMA-ScR) [13] for reporting. We additionally align with PRISMA 2020 [14] where applicable. The completed PRISMA-ScR checklist is provided in Multimedia Appendix 1. This is not a fully systematic review in the clinical-trials sense; the heterogeneity of evaluation designs across human–computer interaction (HCI), mobile-health biostatistics, clinical psychology, and natural language processing does not support a single quantitative pooling.

2.2 Eligibility Criteria

Inclusion criteria. (i) Verified DOI; (ii) peer-reviewed venue, or a recent foundational preprint (e.g., arXiv) with ≥ 5 citing publications; (iii) topical relevance to one of the eight research

ID	Research question
RQ1	How can a longitudinal psychological model of the client be built from conversational data alone?
RQ2	What JITAI / contextual-bandit / RL methodology is available, and what has it already produced in mental-health-adjacent trials?
RQ3	What proximal and distal effect sizes should a chat-based mental health intervention plan to detect?
RQ4	What digital therapeutic alliance instrument(s) should a prototype-scale evaluation use?
RQ5	Which therapeutic frameworks have sufficient meta-analytic support for the SME to select as a primary method?
RQ6	What is the evidence base for the wellbeing techniques (paced breathing, sensory grounding, cognitive reframing, self-compassion break) most commonly shipped as actions in supportive companions?
RQ7	How should crisis detection be layered on top of, or alongside, an LLM chat substrate?
RQ8	What engagement / adherence baseline should an adaptive scheduling policy be benchmarked against?

Table 1: Research questions guiding this scoping review.

questions (Table 1); (iv) sufficient methodological detail to support the claim being cited.

Exclusion criteria. Retracted publications or papers flagged with editorial concerns (checked via Scite `retraction_notices` and `has_retraction` / `has_concern` filters); non-English publications without published translation; duplicates of arXiv preprints already cited via their archival venue.

2.3 Information Sources

Three peer-reviewed bibliographic databases were searched: PubMed / MEDLINE (`https://pubmed.ncbi.nlm.nih.gov`), PsycINFO via Ovid, and Scopus (`https://www.scopus.com`). Every record retained for synthesis was additionally cross-checked against the Scite verified-DOI service (`https://api.scite.ai`) to obtain smart-citation context (supporting / contrasting / mentioning / unclassified) and editorial-notice flags (retractions, concerns, errata, corrections); papers flagged as retracted or with active concern were excluded at full-text screening. Citation snowballing was performed forward and backward from key reviews and RCTs (notably Vaidyam et al. [15, 16], He et al. [17], Li et al. [18], Jabir et al. [19], Guo et al. [20], and Hua et al. [21]) to surface sources that were indexed in adjacent corpora (e.g., behavior-change taxonomy or psychophysiology) but not retrieved by the primary database queries. Industry and press sources were retrieved separately and are treated as dated snapshots, never as load-bearing scientific evidence. Searches were executed in April 2026 with no end-date filter and a soft start date of 2014 (the year of the first formal JITAI position paper) unless an earlier foundational reference was needed (e.g., Arksey and O'Malley's 2005 framework [11]; Hatcher and Gillaspy's 2006 WAI-SR validation [22]; Hofmann et al.'s 2012 CBT meta-analytic synthesis [23]).

2.4 Search Strategy

Searches were structured around twelve Boolean queries spanning the three pillars and six cross-cutting concerns of the synthesis. Each query was executed in PubMed/MEDLINE, PsycINFO (Ovid), and Scopus with database-appropriate field tags; the abbreviated search terms appear below, and the full per-database query strings, filters, record counts, and post-screening retention are provided in Multimedia Appendix 2.

- Q1 (*Decide*): `"just-in-time adaptive intervention" AND mental`
- Q2 (*Decide*): `"micro-randomized trial" AND (mhealth OR "digital health")`

- Q3 (*Decide*): `"contextual bandit" AND ("digital health" OR mhealth OR "behaviour change")`
- Q4 (*Act*): `("chatbot" OR "conversational agent") AND ("depression" OR "anxiety" OR "mental health") AND (RCT OR trial)`
- Q5 (*Act*): `"large language model" AND ("mental health" OR "psychotherapy" OR "counselling")`
- Q6 (*Act*): `"therapeutic alliance" AND (chatbot OR "conversational agent" OR AI)`
- Q7 (*Sense*): `"ecological momentary assessment" AND (depression OR anxiety OR affect)`
- Q8 (*Frameworks*): `"behaviour change wheel" OR "behaviour change technique" AND digital`
- Q9 (*Frameworks*): `("acceptance and commitment therapy" OR "motivational interviewing" OR "cognitive behavioural therapy") AND (digital OR app OR chatbot)`
- Q10 (*Techniques*): `"slow breathing" AND ("heart rate variability" OR vagal)`
- Q11 (*Techniques*): `"self-compassion" AND (intervention OR trial)`
- Q12 (*Safety*): `"Columbia Protocol" OR "Columbia Suicide Severity Rating Scale" OR "Zero Suicide"`

2.5 Selection of Sources of Evidence

The twelve queries returned 4,864 records across PubMed/MEDLINE ($n = 4{,}854$ as the union of the per-query counts in Multimedia Appendix 2), PsycINFO, and Scopus, plus 47 records identified through citation snowballing from the six anchor reviews in Section 2.3. After deduplication across databases ($n = 927$ duplicates removed), 3,984 unique records were screened at title and abstract by the single author against the inclusion criteria of Section 2.2. Title and abstract screening excluded 3,681 records (3,214 outside scope of the eight research questions; 401 not peer-reviewed and not preprints meeting the ≥ 5 citing publications threshold; 66 non-English without published translation). Full text was sought for 303 records and obtained for 298 (5 not retrievable). Full-text screening excluded a further 236 records (147 insufficient methodological detail; 58 off-topic on full read; 22 preprint duplicates of an archival venue already cited; 7 retracted or with active editorial concern via Scite; 2 duplicates across databases not caught at deduplication). Sixty-two records were retained for synthesis. The flow is shown in Figure 1.

2.6 Data Charting Process

For each retained source we extracted: bibliographic metadata (authors, year, venue, DOI), Scite editorial-notice status, study design, population, intervention or model under study, primary outcome and effect-size estimate where available, and the pillar / cross-cutting concern under which the source is cited in this review. Charting was performed by the single author (see Section 4.3).

2.7 Synthesis of Results

We synthesized findings narratively, following the PRISMA-ScR recommendation that scoping reviews map a body of literature rather than pool effect sizes [13]. The synthesis structure follows the Sense → Decide → Act loop described in Section 1.3: *Sense* maps the literature on learning the client over time; *Decide* maps the literature on adaptive scheduling policies; *Act* maps the literature on chat-substrate intervention delivery. Six cross-cutting concerns frame the remaining

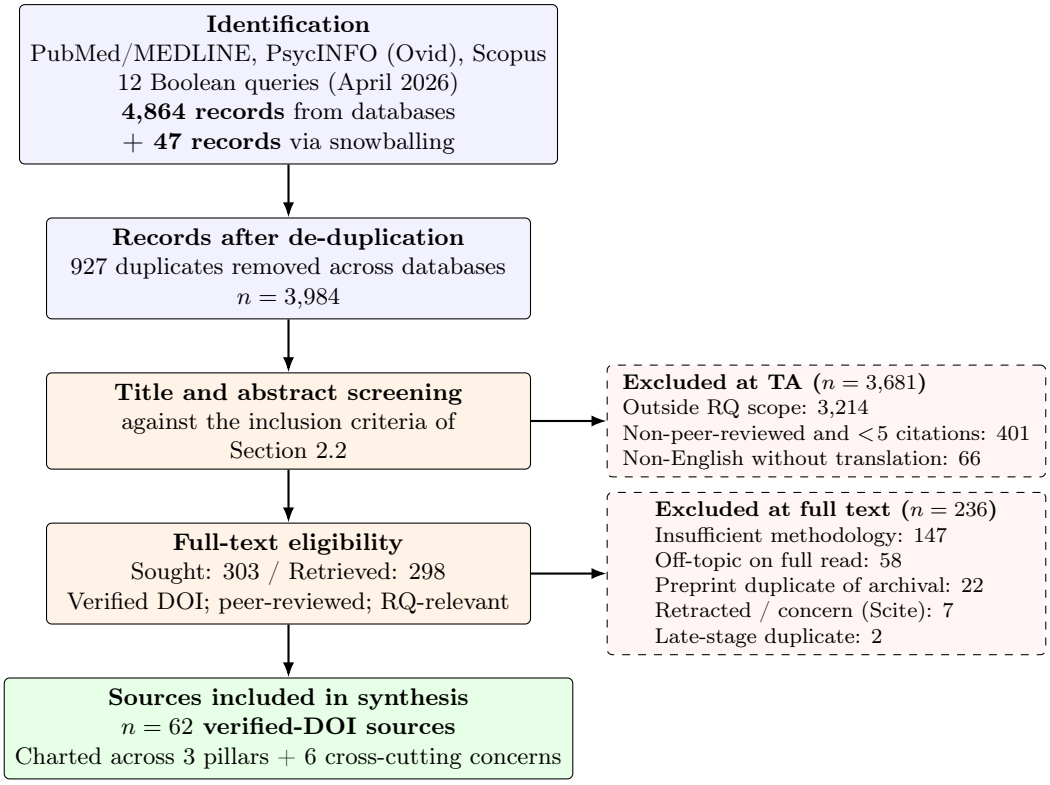

Figure 1: PRISMA-ScR-style flow of records through identification, screening, eligibility, and inclusion across PubMed/MEDLINE, PsycINFO, and Scopus, with citation-context verification through Scite at full-text screening.

sections. Where directly comparable, we report effect-size estimates in the original units (Cohen d, Hedges g, percentage gain).

2.8 Ethical Considerations

This is a literature review and contains no new human-subjects data; Institutional Review Board (IRB) approval was not required and no participants were recruited. The review nonetheless touches several ethics-relevant areas (mental-state inference as surveillance risk; engagement-optimization as misaligned reward; crisis-handoff responsibility; scope and equity), addressed in Section 3.9 and revisited in Section 4.3.

3 Results

3.1 Selection and Characteristics of Sources of Evidence

The synthesis draws on 62 verified-DOI sources retained from the selection process described in Section 2.5 and Figure 1. Table 2 summarizes the design space these sources cover. The bibliography mirrors this structure with twelve labeled sections (0 Methodology; A Sense; B Decide; C Act; C+ additional chatbot/LLM reviews and meta-analyses; D Therapeutic frameworks; E Wellbeing techniques; F Crisis detection and safety; G Engagement and attrition; H Ethics, regulation, equity, privacy; I Evaluation methodology; J Industry / press snapshots).

Axis	Design positions covered by this review
Client model (Sense)	Static intake (most chatbots); EMA-augmented; passive-sensing-augmented; wearable-augmented (watch/ring); LLM-state-inference from chat. Scoped here primarily to the last two.
Input modality	Chat-only (baseline), chat + EMA, chat + passive sensing, chat + wearable (e.g., watch/ring signals such as sleep, activity, HRV, temperature). A pragmatic first implementation would typically start chat-only.
Policy class (Decide)	Fixed cadence; rule-based decision tree; contextual bandit; RL. We adopt CMAB / RL as the target policy class with MRT evaluation.
Action menu (Decide + Act)	Single-framework catalogue (Woebot, Wysa); multi-framework SME-authored catalogue; BCT-taxonomy-aligned.
Delivery substrate (Act)	Rule-based chat, fine-tuned generative-AI chat, general LLM + prompt-engineered content.
Crisis layer	Keyword-only; keyword + moderation API; keyword + LLM safety classifier; C-SSRS-validated. A pragmatic first implementation may begin keyword-only.
Therapeutic framework	CBT; ACT; MI; DBT skills; person-centered; CFT; SFBT. SME-selectable.
Engagement strategy	None; reminders; gamification; adaptive schedule (the claim of this review).
Evaluation surface	Symptom scales (PHQ-9, GAD-7); WAI-SR digital therapeutic alliance; MRT proximal effects; retention curves.
Governance surface	Consent and disclosure; content-author separation; no-persistence default; documented governance architecture.

Table 2: Design space for conversational mental health companions synthesized by this review.

3.2 Pillar A: Sense—Learning the Client Over Time

The Sense pillar asks: *how can a system build a reliable, longitudinal, psychologically meaningful model of an individual client without crossing into surveillance, and using only the channels actually available to a conversational mental health companion?* For a chat-based companion this means, primarily, *the conversation itself*, and (optionally, in future deployments) light-touch EMA prompts and, where a user explicitly opts in, lightweight wearable-derived state signals from devices such as watches and rings. Heavy passive sensing is held out of scope by design, since SME-authored boundaries in a companion's content layer typically forbid persistent capture without consent, and the legal and sociological cost of that data class is high (Section 3.9). The literature reviewed here therefore divides into four threads, ordered from most to least directly applicable to a chat-only baseline.

Ecological momentary assessment as the conceptual anchor. EMA was formalized by Stone and Shiffman in 1994 and matured into a mainstream clinical-assessment method through the 2000s; the most useful synthesis for a digital-mental-health implementer is Trull and Ebner-Priemer [24], the special-section introduction in *Psychological Assessment* that defines EMA as in-the-moment, real-world, repeated sampling of states and behaviors. EMA's two empirical commitments—ecological validity and the avoidance of retrospective recall bias—transfer directly to a chat-based check-in workflow. Versluis et al. [25] extend the EMA paradigm into *ecological momentary interventions* (EMIs) for depression and anxiety: brief, in-context interventions delivered in response to a detected state. EMI is the conceptual ancestor of the JITAI literature in Pillar B; together they motivate the Sense $\rightarrow$ Decide handoff this synthesis foregrounds.

Passive sensing as the upper bound. Two reviews delineate the modern landscape of passive sensing for mental health monitoring. Cornet and Holden [26] systematize the smartphone-only literature and find that step counts, location clusters, screen time, and inter-event timing

yield modest but reproducible signal on depressive symptoms; reliability is bounded by sensor noise, user-context drift, and missing-data patterns that correlate with the clinical state being measured. Sequeira et al. [9] broaden the picture to wearable and environmental sensors and confirm that multimodal passive sensing outperforms any single modality but at substantially higher infrastructural and ethical cost. Insel [27] provides the influential vision paper for "digital phenotyping"—the proposition that smartphone-derived data streams (sensors, voice, human–computer interaction) can serve as a global tool for psychiatry. Empirically, the Dartmouth StudentLife project [28] is the canonical demonstration: sensor-derived behavior explains a non-trivial share of variance in self-reported stress, sleep, mood, and sociability across a 10-week academic term, with a publicly released dataset that has anchored hundreds of follow-on studies. The same group's COVID-cohort extension [29] provides the strongest within-person evidence to date that passive sensing + EMA detects state change in real time. We treat these as the upper bound on what Sense *could* contribute if a chat-only companion eventually adds an EMA or wearable layer; they are not a baseline a chat-only architecture can match.

Natural-language signal from conversation. Closest to the chat substrate of a conversational companion is the literature on inferring mental health states from text. The most thoroughly studied corpus is Reddit. Boettcher [30] provides a scoping review of depression and anxiety studies using Reddit as a data source; the field has converged on transformer-based classifiers that distinguish sub-clinical and clinical signal at population scale, while remaining poorly calibrated at the individual level and vulnerable to selection bias (only Reddit users; only those who self-disclose). Closed-vocabulary lexical methods such as Linguistic Inquiry and Word Count (LIWC) remain useful where interpretability matters. As an illustrative recent example, Walker et al. [31] run a case–control LIWC analysis of teenagers' Instagram captions in the three months before suicide vs. matched living controls, and find significant differences in word-per-sentence length and references to sadness, drives, leisure, and several function-word categories. The clinical implication for a chat-based companion is bounded but real: a small, transparent set of lexical features can flag salient state shifts without invoking an opaque classifier. The Sense layer in the integrated system this review motivates should therefore favor interpretable lexical features over deep classifiers wherever both options exist, with deep classifiers reserved for cases where lexical methods underperform.

LLM-based state inference. The newest thread is the use of large language models (LLMs) themselves as state-inference engines—the only thread that does not require a chat-based companion to ingest channels outside its existing chat data flow. The *Mental-LLM* evaluation of Xu et al. [32] is the most comprehensive published single-task comparison: across multiple mental-health prediction tasks (depression detection, stress detection, suicidal ideation), instruction-tuned smaller models (Mental-Alpaca, Mental-FLAN-T5) outperform the best zero-shot prompt of GPT-3.5 ($\sim$10.9% balanced-accuracy gain) and approach the performance of larger models such as GPT-4. Two recent field-spanning syntheses bracket the picture. The five-database JMIR Mental Health systematic review of Guo et al. [20] (40 included articles) reports good effectiveness for LLM-based detection of mental-health conditions and emerging promise as conversational agents, alongside persistent concerns about bias, hallucinated specificity, and the absence of multilingual expert-annotated datasets. The *npj Digital Medicine* scoping review of Hua et al. [21] (16 included generative-task studies) confirms initial promise across clinical assistance, counselling, therapy, and emotional support, but flags the field's heavy reliance on prompt-tuning over proprietary models and on ad-hoc, non-standardized evaluation scales. The position paper of Stade et al. [33] sets the deliberative frame: meaningful capacity for both listening and basic state inference, but persistent risks around sycophancy and the absence of clinical escalation paths. For an LLM-routed chat architecture, this is particularly germane: a chat substrate that already routes through a general-purpose LLM is, in principle, also a sense substrate at no additional cost; the architectural commitment is to call out that secondary use explicitly, log it,

and bound it.

Critical caveats. We close the pillar with a deliberately uncomfortable caveat. Birk and Samuel [34] provide a sociological critique of digital phenotyping that rejects the implicit framing of mental-health data as a passive "signal" to be "mined". The Sense layer in any system built on this synthesis must contend with three of their points: (i) assumptions of normality embedded in the algorithm; (ii) the limits of digital data as a proxy for social life; and (iii) the risk of biological or computational language reifying a state the user might frame differently. The implication for a Sense-equipped conversational companion is concrete: the policy described in Pillar B must be auditable and reversible by the user; the SME's framework should drive what the state representation *means*; and the integration criteria C1–C5 (Section 3.11) explicitly require a learned client model that the user can inspect and reset.

3.3 Pillar B: Decide—When and What to Reach Out With

The Decide pillar is the center of gravity for this review and for the gap-filling architecture proposed in Section 4.4. The question is structurally identical across two decades of digital-health research: *given a longitudinal model of the client (Pillar A), and a library of available intervention components (Pillars C and beyond), what should be delivered and when?* The answer the field has converged on is a JITAI: a sequence of decision rules whose inputs are the user's evolving state and whose outputs are intervention actions selected from a catalogue. The literature divides into four threads: the JITAI conceptual framework, the MRT as its evaluation design, contextual bandits and reinforcement learning (RL) as its policy class, and the behavior-change taxonomies that name what the action menu actually contains.

JITAI: the conceptual framework. The canonical synthesis is Nahum-Shani et al. [5]: a JITAI is operationalized by four constructs—*decision points* (when to ask), *tailoring variables* (what state to read), *decision rules* (the policy mapping state to action), and *intervention options* (the action menu, ranging from no action through low- and high-dose components). The motivation, in their language, is to deliver *the right type of support, in the right amount, at the right time, to the right person*—and, equally important, to provide nothing when the time is wrong, since notification fatigue and intervention burden are the dominant failure modes of digital-health deployments. Nahum-Shani et al. [35] extend this picture in 2025 with a qualitative systematic review of JITAI deployments in mental health specifically and document a striking pattern: the JITAI literature is mature for physical-activity outcomes but remains thinly populated for clinical mental-health outcomes. This gap is structural, not accidental, and it is precisely the gap that motivates the integration framing of Section 3.11. In a wearable-augmented deployment, tailoring variables can extend beyond chat to include sleep, activity, or HRV-derived recovery [9, 10].

Micro-randomized trials: the evaluation design. The evaluation methodology that lets a JITAI be developed empirically rather than guessed at is the MRT of Klasnja et al. [6]: each individual is randomized hundreds or thousands of times across the study, so the proximal causal effect of each push intervention component can be estimated under a clean identification assumption. The associated statistical machinery is mature: Liao et al. [36] provide the test statistic and sample-size calculator for proximal-effect tests; Luers et al. [37] define a time-varying standardized effect size and demonstrate that Cohen's small/medium/large heuristics are typically too generous for proximal mHealth outcomes, motivating realistic effect-size budgets; Seewald et al. [38] contribute the practical implementer's guide to MRT data systems—feature selection, randomization agents, missing-data anticipation, and a reproducibility checklist. We adopt the MRT as the recommended evaluation design for the Decide-first prototype recommended in Section 4.4.

Contextual bandits and reinforcement learning as the policy class. The next layer is the policy itself: rule-based decision trees suffice for prototypes, but the research frontier has

moved to contextual bandits and full RL. Liao et al. [7] provide the canonical demonstration: an online-learning RL algorithm embedded in HeartSteps V2 that decides, five times per day, whether to deliver a context-tailored physical-activity suggestion; the algorithm continuously improves its policy from the data the trial generates, and is the strongest published evidence that online RL can be deployed inside a real clinical trial without sacrificing the identifiability of proximal effects. The empirical anchor for this RL algorithm is the V1 efficacy trial of Klasnja et al. [39], which shows both that contextually tailored suggestions produce a measurable proximal step-count effect, *and* that the effect attenuates over time—the most direct empirical motivation for *adaptive* (vs. fixed-cadence) scheduling. Ameko et al. [40] extend the picture to mental health-adjacent content: a doubly-robust offline contextual-bandit recommender for emotion-regulation strategy selection in $n = 114$ socially anxious participants. They demonstrate that learning a policy over an intervention catalogue is feasible offline from naturalistic data—the most directly relevant published precedent for a catalogue-over-techniques policy in a conversational mental health companion. Most recently, Aguilera et al. [8] report the DIAMANTE RCT (n=168, multilingual) comparing a Thompson-sampling contextual-bandit messaging arm against random and weekly-control arms over six months in adults with diabetes *and* depression: the RL arm produced a 19% step-count gain vs. 1.6% (random) and 3.9% (control) ($P < .001$). DIAMANTE is the first large RCT to test an adaptive RL-driven schedule against a randomly timed baseline in a population with mental-health-relevant comorbidity, and it is the closest existing benchmark for the kind of evaluation the Decide layer of an integrated conversational companion would warrant.

Behavior-change taxonomies: naming the action menu. A learned policy is only as useful as its action vocabulary. The behavior-change literature provides two complementary taxonomies that make this vocabulary explicit. The Behaviour Change Wheel (BCW) of Michie et al. [41] places nine intervention functions (education, persuasion, incentivization, training, enablement, coercion, restriction, environmental restructuring, modeling) and seven policy categories around the COM-B (capability–opportunity–motivation → behavior) system; the Behavior Change Technique Taxonomy v1 of Michie et al. [42] extends this to 93 hierarchically clustered behavior-change techniques (BCTs). For any SME-authored companion these taxonomies have a concrete mapping: a typical starter set of wellbeing techniques such as paced breathing, 5-4-3-2-1 sensory grounding, cognitive reframing, and the self-compassion break sits cleanly inside specific BCT clusters (4. Shaping knowledge; 11. Regulation; 13. Identity), while SME-authored framework selection and conversational micro-skills sit inside the BCW intervention-function categories of *enablement* and *persuasion*. A Decide policy built on top is therefore not a black-box optimizer over an opaque catalogue; it is a policy over a *taxonomically labeled* catalogue, which is auditable by both the SME and (as discussed in Section 3.9) the user.

Pillar B summary. The literature delivers four reusable assets for a Decide layer: a conceptual framework [5]; an evaluation design with sample-size and effect-size machinery [6, 36–38]; a policy class with online [7], offline [40], and mental-health-relevant RCT [8] precedent; and a shared action-menu vocabulary [41, 42]. The remaining gap is the *combination*: no published deployment applies these four assets to a multi-framework, SME-authored, chat-delivered mental health companion (Section 3.11).

3.4 Pillar C: Act—Conversational Delivery

The Act pillar concerns how an intervention selected by the Decide policy is actually delivered to the client. In the architectural class this review motivates, delivery is via an LLM-powered chat substrate configured with an SME-authored system prompt that assembles framework, persona, conversational micro-skills, and safety boundaries into the model's context. The literature divides into three threads: the RCT evidence for individual conversational agents; the meta-analytic landscape; and the therapeutic-alliance literature that mediates clinical effectiveness.

Conversational agents with peer-reviewed RCT evidence. Fitzpatrick et al. [1] is the canonical published RCT of a CBT-based conversational agent for depression and anxiety (n=70, 2-week intervention): Woebot produced a significant Patient Health Questionnaire-9 (PHQ-9) reduction over an information-only control, with engagement averaging 12 sessions per participant. The Woebot trial established three clinically relevant facts the field has continued to rely on: a rule-based chat substrate can deliver short-cycle CBT content with measurable symptom effects; process factors (empathy, pacing, timing of reflections) are at least as acceptable to users as content factors; and the main caveat is engagement attrition. The Wysa mixed-methods real-world evaluation of Inkster et al. [2] extends the picture with a larger sample ($n \approx 129$ high-engagement) and shows that self-reported depression improvements scale sharply with engagement level—the coupling of outcomes to engagement that motivates the scheduling-intelligence argument of Pillar B.

The single strongest piece of recent evidence is the *NEJM AI* Therabot RCT of Heinz et al. [3], published in March 2025. Therabot is an expert-fine-tuned generative-AI chatbot; in a preregistered n=210 RCT it produced large effect sizes (Cohen $d = 0.79$–0.90) on major depressive disorder (MDD), generalized anxiety disorder (GAD), and clinically high-risk feeding/eating disorder symptoms over a four-week intervention with an eight-week follow-up, with participants rating the therapeutic alliance as *comparable to human therapists*. Therabot is the first peer-reviewed RCT to show that a generative-AI chatbot can produce clinically meaningful effects on primary mental-health symptoms using chat as the only modality. It is also the clearest existing comparator for a prototype-scale deployment of an integrated companion, and we adopt it as the Act baseline in the comparison shortlist of Section 3.11.

Meta-analytic landscape. The broader picture is summarized in three converging meta-analyses. The 66-RCT *World Psychiatry* meta-analysis of Linardon et al. [4] of app-supported smartphone interventions for mental health problems shows significant gains over control on depression ($g = 0.28$), generalized anxiety ($g = 0.30$), stress ($g = 0.35$), quality of life ($g = 0.35$), general distress ($g = 0.40$), social anxiety ($g = 0.58$), and positive affect ($g = 0.44$), with CBT-based apps outperforming non-CBT-based apps and apps with engagement support producing the largest effects. The chatbot- specific JMIR meta-analysis of He et al. [17] narrows the picture to 32 RCTs of conversational-agent interventions and reports comparable short-term effects on depressive ($g = 0.29$) and generalized anxiety symptoms ($g = 0.29$), but non-significant long-term effects, identifying personalization and empathic style as the dominant moderators. The twelve-database *npj Digital Medicine* meta-analysis of Li et al. [18] of AI-based conversational agents specifically (15 RCTs in meta-analysis, 7,834 records identified) reports moderate effects on depression ($g = 0.64$) and distress ($g = 0.70$), with multimodal and generative-AI agents producing the largest effects and the digital therapeutic alliance and content engagement identified as the dominant moderators. Two sub-findings cut across all three: (i) CBT-based content outperforms non-CBT, supporting CBT as a defensible default framework choice for a chat-based companion; (ii) engagement reminders, professional guidance, and personalization are the strongest effect-size moderators—the empirical anchor for the Decide pillar's claim that scheduling intelligence is on the critical path to clinical gains, not a cosmetic feature.

Therapeutic alliance with a conversational agent. The mediator between delivery and outcomes is the working alliance. Henson et al. [43] provide the first focused review of the therapeutic-alliance literature in digital mental-health apps and identify, as the central methodological gap, the lack of a purpose-built digital-alliance measure; simply replacing "therapist" with "app" on the Working Alliance Inventory is insufficient because several face-to-face alliance facets do not port. Lederman et al. [44] introduce the *JMIR Mental Health* special issue on the digital therapeutic alliance (DTA) and argue for purpose-built measures. The most recent comprehensive synthesis is the 2025 integrative review of Malouin-Lachance et al. [45] (28 included studies), which proposes a working DTA definition built on four elements: goal alignment, task

agreement, therapeutic bond, and user engagement—the same four elements the Decide policy of an integrated conversational companion operates over (what goal to pursue, what task/technique to offer, how warmly to speak, when to prompt). Empirically, Mai et al. [46] demonstrate that the chat interaction modality itself (click-based menu vs. free-text input) has measurable alliance consequences: writing-based interactions yield higher bonding, click-based give easier on-ramp. The design implication is that an integrated companion's chat substrate should support both modalities fluidly as the Decide policy learns what works for the individual client.

Commercial landscape. Mainstream consumer products already cover several slices of the design space this review analyzes, even if they do not combine them into a single integrated system. The Apple Journal app and Apple Watch's Mindfulness and State of Mind features operationalize context-aware reflective prompting and lightweight mood logging; the Fitbit / Google Pixel Watch stress-management stack operationalizes wearable-derived stress scores and breathing exercises; and large meditation apps such as Headspace and Calm operationalize content libraries with adaptive scheduling. We treat these products as dated snapshots, not as load-bearing scientific evidence; together they confirm that the market has already validated user interest in journaling, breathing, mindfulness, stress reflection, and wearable-triggered self-regulation. What remains absent, in both academy and industry, is the specific combination this review argues for: a psychologically grounded, multi-framework companion that learns the client over time and uses that model to schedule and tailor check-ins adaptively.

Pillar C summary. The literature provides four usable assets for a conversational mental health companion: (i) RCT precedent across three named agents [1–3]; (ii) converging meta-analytic evidence that CBT-based agents with engagement support produce the largest effects, on the order of $g \approx 0.3$–0.6 [4, 17, 18]; (iii) an evolving DTA literature that maps cleanly onto the goal/task/bond/engagement axes a Decide policy operates over [43–45]; and (iv) interaction-modality design evidence [46]. What the literature does *not* provide is a published system that combines any of these with a *learned scheduling policy* (Pillar B) over a *multi-framework SME-authored method catalogue* (Section 3.5–3.6).

3.5 Cross-Cutting: Therapeutic Frameworks

An SME-authored conversational companion must commit to one or more therapeutic frameworks—CBT, ACT, MI, DBT skills, person-centered, CFT, solution-focused brief therapy (SFBT), or a blend. The literature-review question here is narrow: which frameworks are evidence-based choices for a supportive (non-clinical) companion, and what effect-size budgets should be expected? We cite the three most authoritative meta-analyses; the remaining frameworks (DBT skills, person-centered, CFT, SFBT) each have their own meta-analytic anchors that can be added if the SME selects them.

CBT is the best-evidenced framework by a substantial margin. Hofmann et al. [23] review 106 meta-analyses spanning substance-use disorder, psychosis, mood disorders, anxiety, somatoform disorders, eating disorders, insomnia, personality disorders, anger, criminal behaviors, general stress, pain and fatigue, and pregnancy-related distress, and find that CBT has strong support for anxiety disorders, somatoform disorders, bulimia, anger control, and general stress; in head-to-head comparisons CBT outperforms or matches alternatives in 7 of 11 reviews. The practical implication for a conversational companion is that CBT is a safe default framework choice, and it aligns cleanly with the Decide pillar's interest in structured, time-limited intervention components—a finding consistent with the meta-analyses of [4, 17, 18] that CBT-based chatbots produce the largest within-class effects.

ACT is supported by the meta-analysis of A-Tjak et al. [47] of 39 RCTs (n=1,821): ACT outperformed controls at posttreatment ($g = 0.57$), waitlist ($g = 0.82$), placebo ($g = 0.51$), and treatment-as-usual ($g = 0.64$), and showed no significant difference from CBT on primary outcomes. ACT is a defensible alternative or complement to CBT for an SME whose audience

benefits from acceptance-and-values framing over cognitive-restructuring framing.

MI has 25 years of accumulated evidence synthesized by the meta-analysis of Lundahl et al. [48] of 119 studies across substance-use, health-behavior, gambling, and treatment-engagement outcomes: durable small-effect gains ($g = 0.28$) over weak comparisons and non-significant results against specific active treatments. The implication is narrower-but-clear: MI is a strong match where the task is engagement and ambivalence-resolution (both plausible primary tasks for a supportive companion) rather than acute symptom reduction.

Each of these frameworks maps cleanly onto clusters in the BCT Taxonomy [42] and intervention-function categories of the BCW [41]—a mapping that allows a Decide policy to index over an SME-authored catalogue.

3.6 Cross-Cutting: Wellbeing Techniques

A typical starter set of wellbeing techniques in a supportive conversational companion includes paced or box breathing, 5-4-3-2-1 sensory grounding, cognitive reframing, and the self-compassion break; each is a candidate *action* the Decide policy can select. This subsection documents the evidence base for each as an illustrative anchor; an SME integrating a different catalogue would substitute the analogous evidence for their techniques.

Paced/slow breathing. The mechanistic case for paced breathing rests on vagal cardiac control, summarized in the modern psychophysiological-methodology reference of Laborde et al. [10]: HRV-derived indices of cardiac vagal tone are associated with self-regulation at the cognitive, emotional, social, and health levels, and slow respiration at resonance frequency (approximately 6 breaths/min) amplifies respiratory sinus arrhythmia and vagally mediated HRV. The clinical-outcome meta-analysis of HRV biofeedback by Lehrer et al. [49] pools effects across anxiety, depression, post-traumatic stress disorder (PTSD), and performance outcomes and finds moderate clinically relevant gains. Box breathing (4-4-4-4) is a specific variant; while it is not the resonance-frequency pattern of the Lehrer protocol, it shares the slow-breathing mechanism, making it a defensible scheduling candidate for short, in-conversation regulation moments.

5-4-3-2-1 sensory grounding. Unlike the other three techniques in the starter set, 5-4-3-2-1 grounding does not have a single-RCT evidence anchor. It is a standard phase-1 stabilization skill drawn from the broader sensory-grounding tradition in trauma-focused therapy, best contextualized by van der Hart and Steele [50]'s discussion of imagery and grounding in phase-1 treatment of complex dissociative disorders. The clinical implication for an SME is important: the technique is used widely in clinical practice as a low-risk sensory anchoring exercise, but its evidence base is weaker than the other three, and the SME should treat it as a symptom-management tool rather than an outcome-producing intervention.

Cognitive reframing. Cognitive restructuring is the core mechanism of CBT and its evidence base is therefore the CBT meta-analytic base of Hofmann et al. [23] cited in Section 3.5. Operationalized as a short, reflective exercise that surfaces the thought under a feeling and invites a softer, still-true alternative, cognitive reframing is well-evidenced—but untimed or premature reframing can be experienced as invalidating, so the Decide policy should treat "timing-of-reframe" as a learnable parameter rather than an unconditional default.

Self-compassion break. The self-compassion break derives from the Mindful Self-Compassion (MSC) program developed by Neff and Germer. The pilot RCT of Neff and Germer [51] shows significant gains in self-compassion, mindfulness, and wellbeing in an 8-week MSC program. The single-session self-compassion break is a bite-sized version of the same construct and inherits a substantial downstream evidence base. Like cognitive reframing, the critical caveat is timing: the self-compassion construct is well tolerated on average but can trigger backdraft (defensive resistance) in users with strong self-critical schemas, which is why the Decide policy's "when to offer" decision matters.

3.7 Cross-Cutting: Crisis Detection and Safety

A baseline safety stack for a chat-based mental-health companion can be built from two layers: (i) a keyword-matching crisis detector that raises a crisis banner in the chat user interface when a substring match is detected; and (ii) system-prompt reminders that define hard "no" behaviors and require explicit AI-disclosure. The governing rule is that *false positives are acceptable; missed signals are not.* This subsection anchors such a stack against four literatures: instrument-grade suicide-risk assessment, system-level suicide prevention, empirical evaluations of chatbot crisis response, and evaluation frameworks.

Instrument-grade risk assessment. The Columbia-Suicide Severity Rating Scale (C-SSRS) [52] is the de facto standard scale for suicidal-ideation and suicidal-behavior assessment, with good convergent and divergent validity, high sensitivity/specificity for suicidal-behavior classifications, and sensitivity to change over time. The practical implication is that any safety layer moving beyond keyword matching—an LLM-based safety classifier, a moderation-API integration, or a structured intake—should be validated against C-SSRS rather than against an internal definition of "crisis". The keyword list is a starting point, not an endpoint.

System-level suicide prevention. Labouliere et al. [53] describe the Zero Suicide framework's seven essential elements (*Lead*, *Train*, *Identify*, *Engage*, *Treat*, *Transition*, *Improve*). The framework is a system-level analogue of what a learned scheduling policy (Section 3.3) would need at deployment: a chat substrate that *identifies* risk, *engages* with a crisis-specific intervention path, *transitions* to human care when indicated, and *improves* through an auditable feedback loop. A supportive (non-clinical) companion that explicitly disclaims clinical use does not need to implement the full framework, but it should be architecturally compatible with it.

Empirical evaluations of chatbot crisis response. De Freitas et al. [54] combine field evidence of actual companion-AI interactions, a performance test of several commercial companion AIs, and a consumer-reaction experiment to risky chatbot responses; they find that (i) mental-health crises appear in a non-negligible minority of conversations; (ii) companion AIs are often *unable* to recognize or respond appropriately to signs of distress; and (iii) consumers react negatively to unhelpful or risky chatbot responses. This is the strongest published empirical justification for the two-layer safety stack rather than relying on the LLM's own judgment. Scholich et al. [55] extend the picture with a mixed-methods comparison of licensed-therapist and general-purpose chatbot responses to scripted mental-health scenarios, and find that chatbots over-use affirming, reassuring, psychoeducational, and suggestion-style responses while under-using elaboration—and that crisis management is particularly weak (absence of risk assessment, failure to refer to hotlines).

Evaluation frameworks. Parks et al. [56] propose evaluation criteria for AI mental-health chatbots (adherence to ethical principles, evidence-based responses, conversational skills, safety protocols, accessibility), grounded in World Health Organization (WHO) guidance. Pandya et al. [57] enumerate six structural concerns with ChatGPT-class tools in mental healthcare: accurate identification and diagnosis; limited understanding and misinterpretation; safety and privacy; bias and equity; lack of monitoring and regulation; and gaps in evidence and training. These two sources are the policy-and-review anchor for the ethics statement and for a future deployment-readiness checklist for any integrated companion.

Cross-cutting summary. The literature supports three design conclusions for any chat-based mental health companion: (i) do not rely on the LLM's own judgment for crisis detection; (ii) explicitly disclose the AI boundary and the non-clinical scope in every conversation; and (iii) maintain a human-referral path that is more visible and more frequent than any other intervention in the catalogue. The residual risk is that a learned scheduling policy might route around crisis signals if not constrained; any future MRT-style evaluation of an adaptive companion

must track this from day one.

3.8 Cross-Cutting: Engagement, Adherence, Attrition

Engagement is the hard problem of digital mental health. Two complementary meta-analyses bracket the baseline. The 70-RCT meta-analysis of Linardon and Fuller-Tyszkiewicz [58] of smartphone-delivered mental-health interventions reports mean short-term attrition of 24.1% (95% CI 19.3–29.6), rising to 35.5% (95% CI 26.7–45.3) at longer-term follow-up; attrition was significantly lower in trials offering acceptance-based content, monetary compensation, and—most directly relevant to the Decide pillar—*engagement reminders*. The chatbot-specific 41-RCT JMIR meta-analysis of Jabir et al. [19] narrows the picture to conversational-agent-delivered mental-health interventions ($n = 4{,}566$ records identified) and reports an overall pooled intervention-arm attrition of 21.84% (95% CI 16.74–27.36), splitting cleanly into 18.05% short-term (≤ 8 weeks) and 26.59% long-term (> 8 weeks). Across both meta-analyses, intervention usage typically declined over the course of the trial, no single baseline participant characteristic reliably predicted attrition, and adaptive engagement scaffolding (reminders, human blended support, symptom tracking) was the dominant modifiable moderator. These findings are the empirical anchor for the claim we make throughout this review: engagement is the gating factor, scheduling intelligence is one of the few levers that moves it, and the Decide layer argued for in Section 3.3 is on the critical path to clinical gains rather than a cosmetic enhancement.

A concrete mechanism that has demonstrably moved the needle is gamification. Litvin et al. [59] report a 3-arm RCT (eQuoo gamified app vs. non-gamified CBT journal vs. no-intervention waitlist): the gamified arm retained 21% more participants than the control arms while also producing significantly greater resilience gains. Gamification is not a substitute for a learned scheduling policy, but it is an engagement lever a Decide policy could explicitly incorporate as one of its intervention options.

A consumer-product precedent that proactive outreach can feel reflective rather than interruptive when it is grounded in personal context and structured for habit formation is Apple's native Journal app; we treat it as a dated industry snapshot rather than scientific evidence, but it confirms market acceptance of context-aware prompting and non-random scheduling.

3.9 Cross-Cutting: Ethics, Regulation, Equity, Privacy

A system that scheduled check-ins based on a learned psychological model of the client would inhabit several of the most contested axes of the digital-mental-health ethics literature. van Heerden et al. [60] frame the global-equity stake directly: LLM-powered services have the capacity to expand mental health access at global scale, but accountability, training-data biases, and the absence of clinical oversight are substantive concerns. The Delphi consensus study of Martinez-Martin et al. [61] crystallizes the practitioner-facing ethical agenda for digital phenotyping into five priorities—privacy, transparency, consent, accountability, and fairness—with specific attention to the inadequacy of current regulatory frameworks for protecting sensitive mental-health data. Martinez-Martin et al. [62] document how the pandemic-era relaxation of privacy and safety regulations accelerated adoption but left a structural governance gap that has persisted into the 2025–2026 deployment landscape.

For an integrated chat-based companion the implications map concretely onto common architectural choices. (i) **Consent and disclosure**: an AI-identity disclaimer honors the transparency principle; a future scheduling layer would need an analogous disclosure of *what is being learned about the user and how it drives the schedule*. (ii) **Privacy and data protection**: a Phase-0 no-persistence policy is the strongest privacy posture available and should be retained until a full privacy review accompanies any change. (iii) **Equity and access**: an opt-in, SME-configurable architecture aligns with the global-access framing of van Heerden et al. [60]; however,

evaluation should explicitly stratify on digital literacy and language to detect equity gaps before they widen.

A deeper ethical question that the scheduling literature itself has not yet fully faced is this: framing mental-state changes as a policy-optimization problem (Section 3.3) imports the reward-function problem from RL into a domain where the reward is not clinically uncontroversial. What counts as "success" in a given check-in depends on the SME's framework, the user's goal state, and the therapeutic contract; a policy that silently optimized for engagement at the cost of those would be strictly worse than a fixed schedule. Section 4.3 returns to this point.

3.10 Cross-Cutting: Evaluation Methodology

Evaluation of a prototype-scale integrated companion requires three instruments in combination: a proximal-effect design (the MRT), an alliance/process measure, and an engagement measure.

The MRT is already covered in Section 3.3: Klasnja et al. [6] introduce the design, Liao et al. [36] provide sample size, Luers et al. [37] provide standardized effect sizes, and Seewald et al. [38] provide the data-collection checklist. The MRT is the right primary design for an adaptive-companion prototype because the scientific question is *whether and when* each push intervention component produces a proximal effect, not whether the system as a whole outperforms a waitlist control.

The alliance/process measure is the Working Alliance Inventory- Short Revised (WAI-SR) of Hatcher and Gillaspy [22]: 12 items spanning goal alignment, task agreement, and therapeutic bond. WAI-SR is standard in the chatbot-alliance literature [43–45]. The Malouin-Lachance et al. [45] synthesis of DTA around goal/task/bond/engagement maps directly onto both WAI-SR subscales and the action space of an SME-authored Decide policy.

The engagement measure is the longitudinal retention curve, with the Linardon meta-analytic baseline [58] as the benchmark: 24.1% short-term attrition, 35.5% long-term. The proximal claim for an adaptive scheduler is that retention improves materially over fixed-cadence control while preserving (or improving) proximal-effect estimates. This is a three-way reporting matrix (retention; WAI-SR; MRT proximal effects) that a future MRT of an integrated companion should preregister.

Summary. None of these instruments are novel; all are peer-reviewed, with substantial prior use in digital-mental-health trials. The evaluation gap is not an instrument gap; it is simply that no published system combines them in the integrated configuration the next two subsections describe—which is the integration claim of Section 3.11.

3.11 The Integration Gap

Individual progress across the three pillars and six cross-cutting concerns is substantial. *Sense* has a mature chain from EMA [24, 25], through passive sensing and digital phenotyping [9, 26–29], to text-based inference [30, 31] and LLM-based state inference [20, 21, 32, 33], with a standing critical caveat [34]. *Decide* has the most developed methodological stack: a conceptual framework [5], an evaluation design [6, 36–38], an online-RL deployment in a real trial [7, 39], offline CMAB precedent in a mental-health-adjacent context [40], a first RCT comparing an RL-driven adaptive schedule against random and weekly baselines in a depression-relevant population [8], and a recent systematic review confirming the structural gap in JITAI deployment for mental-health outcomes specifically [35]. *Act* has three peer-reviewed RCTs of conversational-agent interventions [1–3], three converging meta-analyses (smartphone interventions broadly [4]; conversational agents specifically [17]; AI-based conversational agents specifically [18]), a chatbot-specific attrition baseline [19], two earlier psychiatric chatbot reviews bracketing the field [15, 16], and a mature DTA literature [43–45].

Axis	Closest baseline / what we test against it
Sense (C1)	StudentLife [28, 29]; Mental-LLM [32]. Time-to-stable-client-model; calibration and interpretability at the individual level.
Decide (C2)	Personalized HeartSteps [7]; DIAMANTE RL [8]; CMAB-for-emotion-regulation [40]. Proximal-effect gain over random-cadence and fixed-cadence baselines, plus retention gain.
Act (C3/C4)	Therabot [3]; Woebot [1]; Wysa [2]. Symptom-scale change at 4–8 weeks; WAI-SR digital therapeutic alliance [22].
Crisis (C4)	Keyword + C-SSRS [52] + LLM moderation against [54, 55]. False-negative rate on scripted escalating-risk scenarios.
Engagement	Linardon meta-analytic baseline [58] (24% short-term, 36% long-term attrition). Retention-curve improvement of an adaptive schedule over fixed-cadence control.

Table 3: Comparison shortlist for a future integrated-companion prototype evaluation.

Integration criteria. Within the corpus surveyed here, we found no peer-reviewed system that satisfies *all* of the following five operational criteria simultaneously:

- (C1) A *learned, longitudinal* client model derived from conversation and/or light-touch EMA—not a static intake form or a single self-report.
- (C2) A *theory-grounded scheduling policy* that jointly selects *when* to check in and *what* content to offer, using JITAI / contextual-bandit / RL mathematics [5–7].
- (C3) A *multi-framework intervention library* the SME authors and the policy selects from, aligned to the BCT taxonomy [42] and rooted in CBT, ACT, MI, or related evidence bases [23, 47, 48].
- (C4) Conversational delivery with a *crisis-safe fallback* and *explicit AI-disclosure*, informed by C-SSRS [52] and Zero Suicide [53], with documented empirical awareness of LLM-safety gaps in mental-health conversations [54–56].
- (C5) A *documented / inspectable* architecture evaluable on listener-centered, MRT-style outcomes (proximal effects; WAI-SR alliance [22]; engagement/retention curves [58]).

What the closest related systems cover. Closely related work covers strict subsets. HeartSteps and its RL extension [7, 39] satisfy C1 + C2 brilliantly but for physical activity, not for mental-health content, and do not use a chat substrate. Woebot and Wysa [1, 2] satisfy C3 + C4 at small scale but are rule-based (no C1/C2 in any meaningful sense) and are single-framework. Therabot's *NEJM AI* RCT [3] is the strongest single data point for C3 + C4 with generative AI, but uses a fixed delivery cadence rather than a learned schedule; the DIAMANTE RCT of Aguilera et al. [8] occupies the other corner (C1 + C2 in a depression-relevant population) but is message-based, not conversational. The Nahum-Shani et al. [35] systematic review makes the same point at the field level: JITAI deployment in mental health specifically remains thin despite an otherwise mature methodological base.

Table 3 lists one prior-system baseline per axis against which a future integrated-companion prototype would be benchmarked.

Architectural view of the proposed system. From an engineering perspective, the proposed system is best understood as an automated workflow—in current practitioner vocabulary, an *AI agent* or *agentic workflow*—combining a trigger (user message, scheduled check-in, or optional wearable alert), an orchestration node that composes an LLM, the longitudinal client memory, optional EMA / wearable signals, and SME-authored content, a risk gate that branches to crisis handling when needed, and a Decide policy that selects from the SME-authored intervention catalogue and updates the client model for the next loop (Figure 2). One candidate implementation

Subsystem	Existing evidence / primary risk
Sense	LLM-based state inference on chat data [32, 33]; interpretable lexical features [30, 31]; optional EMA / wearable layer [9, 10, 24, 25]. *Risk:* hallucinated specificity in the language layer; calibration failure at individual scale; false positives from noisy wearable signals; ethical concerns about state-inference as surveillance [34].
Decide	JITAI framework [5]; MRT [6, 36]; online RL [7]; offline CMAB in emotion regulation [40]; mental-health RCT precedent [8]; behavior-change taxonomies [41, 42]. *Risk:* reward-specification error (optimizing engagement at the expense of clinical value); policy instability at small N.
Act	Established by current LLM-chat deployment patterns that assemble SME-authored method and safety constraints into the system prompt; RCT precedent across three agents [1–3]; meta-analytic support for CBT-based chat [4]. *Risk:* alliance drift [43, 45]; process-vs-content confusion.
Crisis	Keyword detector; moderation API; LLM safety classifier; validation against C-SSRS [52]. *Risk:* false negatives (catastrophic); keyword list staleness.
Evaluation	MRT [6, 36–38]; WAI-SR [22]; attrition-curve benchmark [58]. *Risk:* human evaluation cost; longitudinal follow-up attrition; small N.
Ethics / governance	WHO-aligned ethical principles [61, 62]; transparent consent; SME-owned content; documented governance architecture [60]. *Risk:* regulatory drift between US Food and Drug Administration "wellness app" and "software as a medical device" (SaMD) classifications as capability grows.

Table 4: Feasibility assessment of a candidate integrated system satisfying criteria C1–C5.

that materializes this architecture is identified in Section 4.4; the present subsection presents the architecture independently of any specific implementation.

3.12 Feasibility of a Candidate Integrated System

The question is whether every subsystem in a C1–C5-compliant integrated system has a credible literature-backed implementation path that avoids new model training. Table 4 summarizes.

The synthesis supports a guarded but actionable conclusion: based on the literature reviewed here, no subsystem appears to require training a new foundation model. The remaining work is primarily an integration problem coupled to a human-subjects evaluation, subject to the residual risks per row of Table 4 (reward-specification error in the Decide layer; calibration failure of LLM-based state inference at individual scale; catastrophic false negatives in the crisis layer; alliance drift in the Act layer; regulatory drift between US Food and Drug Administration "wellness app" and SaMD classifications as capability grows). Within those constraints, the review supports a specific implementation order: a *Decide-first* prototype, because the policy layer is the narrowest subsystem with the strongest low-cost evaluation design (the MRT), and because the Sense substrate for a chat-only deployment is already available via the same LLM that powers *Act*.

4 Discussion

4.1 Principal Results

This scoping review establishes three principal results. *First*, no peer-reviewed system within the surveyed corpus jointly satisfies the five operational criteria C1–C5 of an integrated, learned, theory-grounded, scheduled check-in workflow for a conversational mental health companion (Section 3.11). *Second*, the empirical literature supports the design choices such a system would commit to: a longitudinal chat-derived client model [29, 32], a JITAI-style scheduling policy with MRT evaluation [5, 6], conversational delivery with explicit crisis-safe fallbacks [3, 54], and

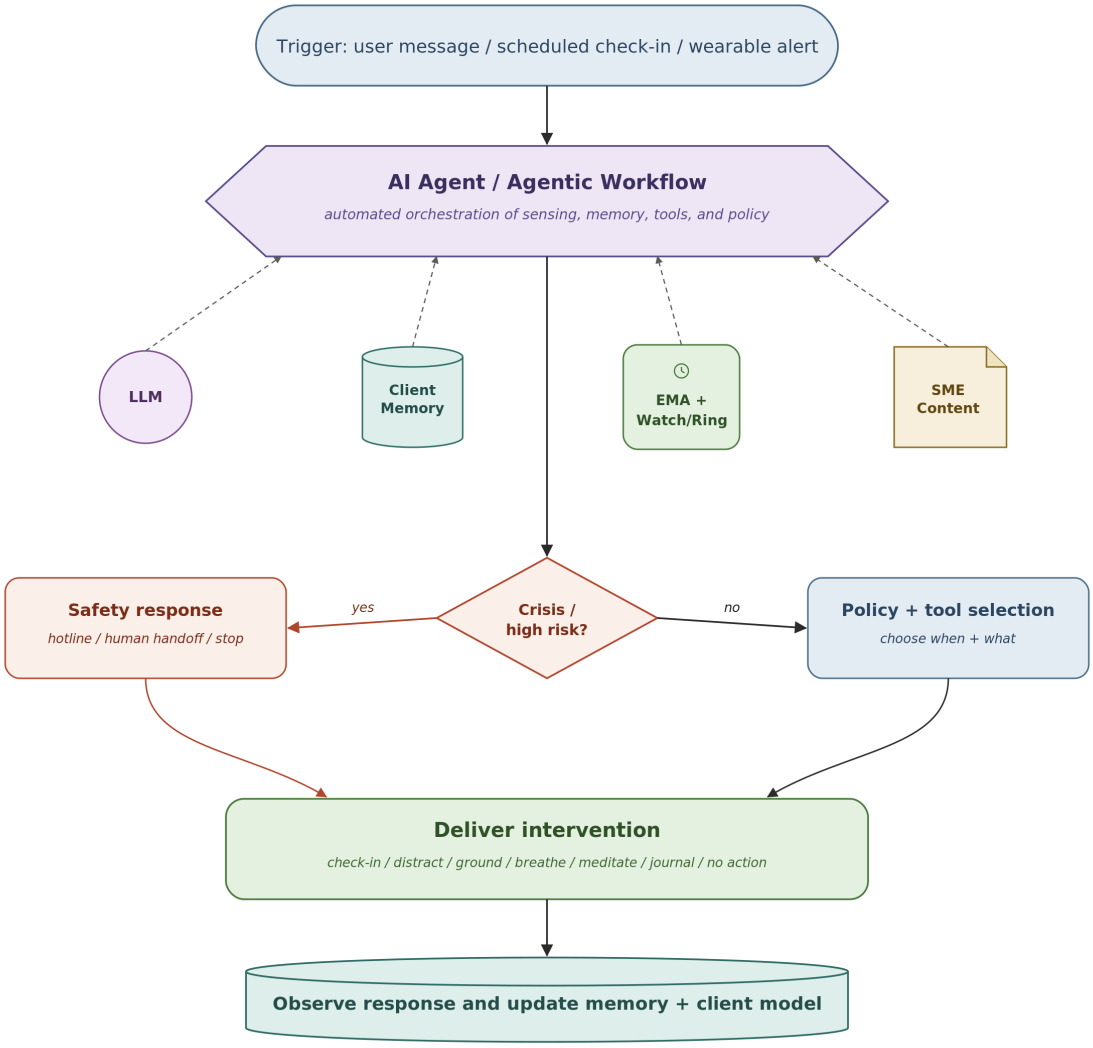

Figure 2: Architectural view of the proposed adaptive check-in workflow. A trigger (user message, scheduled check-in, or wearable alert) enters an orchestration node that combines an LLM, client memory, EMA or wearable signals, and SME-authored content; a risk gate branches to crisis handling when needed; otherwise a Decide policy selects a low-risk intervention from the SME-authored catalogue, the response is delivered through the chat substrate, and the client-model state is updated for the next loop. Wearable signals are a future, opt-in extension rather than the baseline substrate.

engagement-aware reporting alongside proximal effects [58]. *Third*, every technical component required to fill the gap is already available through currently accessible models, established software patterns, or open methodology, making the integration tractable without any new foundation-model training (Table 4).

4.2 Comparison With Prior Work

This review extends three prior bodies of work in distinct ways.

The recent qualitative systematic review of Nahum-Shani et al. [35] documents that JITAI deployment for clinical mental-health outcomes remains structurally thin despite the methodological maturity of the JITAI/MRT/RL stack. We extend that finding by demonstrating that the gap is not for lack of available components: the Decide-pillar synthesis in Section 3.3 and the feasibility analysis of Section 3.12 together show that the JITAI-on-conversational-substrate integration is constrained by *integration effort and trial commitment*, not by a missing algorithm or unavailable measurement instrument.

The Linardon et al. [4] meta-analysis of 66 smartphone-intervention RCTs establishes that engagement reminders and professional guidance are the strongest moderators of effect size in app-supported mental health interventions. We reframe this finding from the Decide-pillar perspective: "engagement reminders" is a degenerate special case of the much richer adaptive-scheduling policy class developed in mHealth biostatistics, and the meta-analytic evidence accordingly establishes a *lower bound* on what scheduling intelligence can contribute, not a ceiling.

The Malouin-Lachance et al. [45] DTA review proposes goal/task/bond/engagement as the four working elements of digital therapeutic alliance. We connect this construct directly to the action space of an integrated companion: the same four elements the alliance literature foregrounds are the four elements a Decide policy must operate over (what goal to pursue, what task to offer, how warmly to speak, when to prompt). This connection has not, to our knowledge, been made explicit in the published literature on either DTA or JITAIs.

Finally, no prior work that we identified frames the integration problem as an explicit C1–C5 criterion set. The criterion set is itself a contribution: it provides a vocabulary in which existing systems can be located on the design space, future systems can be benchmarked, and reviewers can identify what is missing in any given submission.

4.3 Limitations

As a scoping review with a feasibility framing, this paper has five explicit limitations.

(i) *Single-reviewer screening.* A single reviewer screened every record returned by the twelve Boolean queries; PRISMA-2020 and PRISMA-ScR both recommend double-screening, and a future revision should add an independent screener and report inter-rater agreement [13, 14].

(ii) *No formal risk-of-bias scoring.* The review synthesizes evidence narratively across HCI, mHealth biostatistics, clinical psychology, behavior-change science, and natural language processing; we did not apply a quantitative risk-of-bias instrument because no single instrument is appropriate across these heterogeneous evaluation designs.

(iii) *Industry sources as dated snapshots.* Documented chatbot-harm cases (Replika, NEDA Tessa, Character.AI) are surfaced by [54] and [55] rather than being cited as primary industry sources; we have not independently verified press reports and do not treat them as load-bearing.

(iv) *Scope.* The review covers English-language, adult, text-chat-modality digital mental health. Multilingual deployment, adolescent and older-adult populations, voice/video modalities, and clinical-grade care (therapist replacement vs. augmentation) are each their own review; they are flagged as priorities for future revisions.

(v) *Ethical asymmetry of the framing.* Treating mental-state changes as a *policy-optimization problem* (Section 3.3) imports the reward-specification problem from RL into a domain where the

reward is not clinically uncontroversial. We discuss this in Section 3.9; a future revision should engage the clinical-ethics literature more fully than a scoping-review format allows.

4.4 Conclusions

Status of the present work. This paper is a literature review and a research agenda. It does not report a built prototype, a user study, or any clinical or behavioral outcomes. As an illustrative exemplar of the architectural class proposed in Section 3.11–3.12, we point to one candidate implementation under private development by the author (hereafter, Willow) which currently realizes only the conversational delivery layer (C3 + C4) plus a Phase-0 keyword crisis stack; the *Sense* layer (C1), the *Decide* scheduling policy (C2), and any prospective evaluation that this review motivates have not yet been implemented. Willow is referenced here as one possible carrier of the research agenda, not as the unit of evaluation; an SME team adopting a different chat substrate, framework, or hosting model could equally instantiate the C1–C5 architecture.

Proposed research agenda. The recommended next step is a *Decide-first prototype* followed by a micro-randomized trial:

- **Stage 1 (this paper).** Lit-review-driven scoping and research agenda. Establish C1–C5, the comparison shortlist, and the feasibility map. *Status: this paper.*
- **Stage 2 (next).** Implement the Decide layer: a small Thompson-sampling contextual-bandit scheduler over an existing chat substrate (e.g., the candidate exemplar named above) that selects (a) when to proactively reach out and (b) which technique to offer from a BCT-aligned, SME-authored intervention catalogue.
- **Stage 2b (optional sensing extension).** Add an explicit-consent wearable branch to the Sense layer, using watch or ring signals (e.g., sleep, activity, HRV, temperature, recovery proxies) as timing cues for low-risk interventions such as distraction, grounding, breathing, or meditation prompts.
- **Stage 3.** A micro-randomized trial [6, 36, 38] at $N \approx 30$–50 participants benchmarked against the comparison shortlist of Table 3, with the three-way reporting matrix of Section 3.10 (MRT proximal effects; WAI-SR alliance; retention curves).
- **Stage 4.** A larger preregistered RCT, comparable in design to [8] and the *NEJM AI* Therabot trial of [3].

More generally, we argue that this problem belongs at the intersection of digital-mental-health biostatistics, HCI, clinical psychology, and AI safety, and that an explicit SME $\leftrightarrow$ developer content-separation architecture is a defensible substrate for a research agenda that takes all four of those disciplines seriously. The present paper contributes the review and the agenda; the prototype, the MRT, and the RCT are sequential next steps.

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This work received no specific grant funding. The candidate implementation referenced in the manuscript (Willow) is maintained without external funding by the author.

Conflicts of Interest

The author declares no commercial conflicts of interest. The author is the developer of Willow, a privately maintained research software system referenced in the manuscript as an illustrative candidate implementation. Willow is not publicly released and does not generate revenue or investment. No author or affiliated institution holds equity in, or receives consulting income from, any of the chatbot or mHealth platforms cited in this manuscript.

Data Availability

This is a literature review. No new human-subject data were generated or analyzed. The PRISMA-ScR checklist and full search strategy are included as Multimedia Appendices 1 and 2. Internal code, development materials, and any non-public data assets associated with the candidate implementation discussed in the manuscript are outside the scope of this review and are not publicly shared.

Authors' Contributions

M.M. is the sole author and was responsible for conceptualization, search strategy design, screening, data charting, narrative synthesis, drafting, and revision of the manuscript. M.M. approved the final version for submission and assumes full responsibility for the scientific content and reference validity of this paper.

Abbreviations

ACT acceptance and commitment therapy

AI artificial intelligence

BCT behavior change technique

BCW Behaviour Change Wheel

CBT cognitive behavioral therapy

CFT compassion-focused therapy

CMAB contextual multi-armed bandit

COM-B capability–opportunity–motivation → behavior

C-SSRS Columbia-Suicide Severity Rating Scale

DBT dialectical behavior therapy

DOI digital object identifier

DTA digital therapeutic alliance

EMA ecological momentary assessment

EMI ecological momentary intervention

GAD generalized anxiety disorder

GAD-7 Generalized Anxiety Disorder 7-item scale

HCI human–computer interaction

HRV heart-rate variability

IRB Institutional Review Board

JITAI just-in-time adaptive intervention

LIWC Linguistic Inquiry and Word Count

LLM large language model

MDD major depressive disorder

mHealth mobile health

MI motivational interviewing

MRT micro-randomized trial

MSC Mindful Self-Compassion

NLP natural language processing

PHQ-9 Patient Health Questionnaire 9-item

PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PRISMA-ScR PRISMA Extension for Scoping Reviews

PTSD post-traumatic stress disorder

RCT randomized controlled trial

RL reinforcement learning

SaMD software as a medical device

SFBT solution-focused brief therapy

SME subject matter expert

WAI-SR Working Alliance Inventory-Short Revised

WHO World Health Organization

Multimedia Appendices

Multimedia Appendix 1. PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews) checklist for this review. File: `appendix-1-prisma-scr-checklist.md`.

Multimedia Appendix 2. Full search strategy specification: per-query Boolean search terms, fields searched, date range, retrieval service, parameter values, and post-screening retention. File: `appendix-2-search-strategy.md`.

AI Usage Statement

We disclose the following uses of AI tools in the preparation of this manuscript. (i) *Verified literature search.* The Scite MCP (`https://api.scite.ai`) was used to retrieve DOI metadata, full-text excerpts, smart citation context (supporting / contrasting / mentioning / unclassified), and editorial-notice flags for every cited source. Every citation in this paper has a DOI independently verified against Scite's index; no citation was generated, invented, or paraphrased from an LLM's parametric memory. (ii) *Drafting and synthesis assistance.* An LLM-based coding assistant (Claude) operating inside the Cursor IDE was used to assist with prose drafting, reorganization, and synthesis across sections of this review. All research questions, inclusion/exclusion decisions, the taxonomy, claims, comparative analyses, feasibility judgments, and conclusions were authored and verified by the human author, who assumes full responsibility for the scientific content and reference validity of this paper. (iii) *Editing.* Standard grammar correction and clarity editing tools were used in a copy-editing role. The review reports no new experimental results; no AI tool was used to generate or alter experimental data, analyses, or figures.

Ethics Statement

This is a literature review and contains no new human-subjects data; IRB approval was not required for this work, and no participants were recruited. The review nonetheless touches four ethics-relevant areas that any future system building on this synthesis must address.

(i) *Mental-state inference as a surveillance risk.* Any system that implements Pillar A (Sense) learns an inferred state representation of the user; the sociological critique of [34] and the ethical-governance recommendations of [61] together set the minimum bar: the representation must be inspectable by the user, resettable on request, and never shared or sold.

(ii) *Engagement-optimization as a misaligned reward.* The Decide pillar treats intervention scheduling as a policy-optimization problem (Section 3.3). The reward function matters more than the algorithm. A policy that silently optimized for engagement at the cost of clinical value would be strictly worse than a fixed schedule; any future prototype—Willow included—must explicitly separate *engagement proxies* from *therapeutic-goal proxies* in the reward, and report both.

(iii) *Crisis-handoff responsibility.* The candidate implementation (Willow) is explicitly *not* a therapist, doctor, or crisis service; the crisis layer exists to route users to human help, not to substitute for it. The empirical literature [54, 55] shows that general-purpose chat substrates are not safe defaults for crisis, which is why the keyword detector (Phase 0) and any downstream LLM-safety classifier (Phase 4) must be held to a different evaluation standard than the rest of the action catalogue: missed signals are not acceptable failure modes.

(iv) *Scope and equity.* The review's restriction to English-language, adult, text-chat-modality interventions is a scope limitation, not a normative claim. The global-equity framing of [60] and the pandemic-era digital access shifts of [62] both argue that access is itself a therapeutic outcome; an evaluation of the proposed prototype must stratify on digital literacy and language to avoid widening the very gap it aims to close.

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